The Production AI & Agentic Stack

Inspired by today's signal (Decrypt: "Republican Senator Wants to Freeze AI Data Center Boom as Congress Pus") — 5 open-source repos, evaluation labs & checklists to ship reliable AI.

★ BOOKMARK & SAVE FOR YOUR NEXT AI SPRINT

01 High-Throughput Local & Cloud Inference Engines (vLLM & LiteLLM)

02 Deterministic Agent Eval & Red-Teaming Harnesses (Promptfoo)

03 Free Interactive Deep Tech Courses (Hugging Face & DeepLearning.AI)

04 5-Minute Production Agent Readiness Self-Audit Checklist

From Raw Model Size to Systems Alpha

Today's catalyst (Senator Josh Hawley wants a nationwide construction pause until households are protected from rising power bills, even as Congress) underscores a critical shift for engineering leaders:

BOTTLENECK 1

Unbounded Token & Latency Sprawl

Passing raw, unindexed context windows into frontier models inflates inference spend by 10× while degrading multi-step reasoning precision.

BOTTLENECK 2

Vibes-Based Evaluation

Shipping prompt or model updates without automated regression suites leads to silent production hallucinations and schema drift.

THE SOLUTION

Structured Context + Deterministic Evals

Order flow in digital asset markets at 6% reflects tightening market liquidity and higher institutional verification standards.

Promptfoo (Deterministic LLM Evals)

Stop testing prompts by hand. Run automated matrix benchmarks, red-team attacks, and CI/CD assertions locally.

OPEN SOURCE REPO

★ 10,500+ Stars

`github.com/promptfoo/promptfoo`

CLI and library for evaluating LLM outputs, comparing models side-by-side (Claude, GPT, Gemini, local Llama), and catching regressions in CI before merge.

RUN IN 1 COMMAND:

```
npx promptfoo@latest init && npx promptfoo@latest eval
```

💡 **Pro Tip:** Enforce JSON schema & latency assertions on every PR touching agent prompts.

vLLM & LiteLLM Gateway

Unify 100+ model providers behind one OpenAI-compatible API with PagedAttention throughput, fallbacks, and token budgets.

PRODUCTION INFRASTRUCTURE

★ 32,000+ Stars

`github.com/vllm-project/vllm • github.com/BerriAI/litellm`

Decouple your application code from single-vendor lock-in. Route low-complexity tasks to fast distilled models and reserve frontier reasoning for complex synthesis.

- ✓ **PagedAttention:** Up to 24× higher serving throughput via continuous batching.
- ✓ **Zero-Downtime Fallbacks:** Auto-route across providers on HTTP 429/503 errors.
- ✓ **Per-Agent Cost Caps:** Strict virtual keys and token budgets per microservice.

> 💡 **Architecture Law:** Own your routing layer and knowledge ontology; treat LLMs as swappable compute engines.

Hugging Face Agents & DeepLearning.AI

Hands-on, 100% free interactive notebooks and engineering labs to master tool-use, RAG, and multi-agent orchestration.

100% FREE LABS

Interactive Notebooks & Certificates

`huggingface.co/learn/agents-course`

Step-by-step engineering curriculum covering ReAct loops, code-generating agents, memory graphs, and production observability with real open-weight models.

- Tool Calling & Schema Validation
- Low-Token Knowledge Graph RAG
- Multi-Agent Handoffs & Guardrails
- Quantization (AWQ / GGUF / FP8)

> 📖 **Companion Repo:** Bookmark `github.com/anthropics/anthropic-cookbook` for production patterns.

The 5-Minute AI Production Audit

Run these 5 engineering checks across your AI/agent pipeline before your next production release:

- ✓ **Strict Token Budgets:** Compress raw documents into structured knowledge cards (<800 tokens) before prompting.
- ✓ **Schema-Enforced Outputs:** Require strict JSON/typed schemas with deterministic retry/validation gates.
- ✓ **Golden Dataset Evals:** Maintain at least 50 real-world edge cases in CI that run on every prompt/model change.
- ✓ **Least-Privilege Tool Sandboxing:** Never grant autonomous agents unrestricted write/network execution without human-in-the-loop sign-off.
- ✓ **Multi-Provider Fallback:** Configure automatic failover so a single API outage never halts your production workflow.

Bookmark This Swipe-File 🔖

In 2026, durable alpha belongs to teams with verified data ontologies and deterministic harnesses — not raw wrapper prompts.

📌

Save This Post For Later

Tap the **three dots (•••)** at the top right of this post and click '**Save**' to revisit this toolkit during your next architecture review.

DISCUSSION PROMPT:

"Given today's move (Republican Senator Wants to Freeze AI Data Center Boom as Congres), what is the #1 bottleneck in your current AI stack: evals, latency, or context quality?"

Share your architectural perspective in the comments below.

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VERIFIED DISPATCH